Support for the New Claims

The overall architecture recited in new independent claims 19 and 35 is summarized at page 8, lines 9-26, and is described in detail with reference to FIG. 1 at pages 13-15, in particular from page 14, line 2, through page 15. These passages describe the mate creation component 110, the transcoding components 120, the manifest files 121 pointing to unique interleaved combinations of chunks 113 of the ensemble 112, the ledger 122, the detection component 130 running a camera cuts detection algorithm 131 that devises time codes from the pirate webcasting and builds reconstructed manifest files 121′, and the retrieval component 140 searching the ledger to identify the associated spectator or group. The preferred manifest embodiment states that the retrieval component searches for delivered manifest files equal to the reconstructed manifest files 121′. Original claims 1, 10, 11, 14 and 15 recite this architecture in claim form. Original claim 1 identifies a distinguishable version matching the suspected unauthorized distribution and original claim 15 recites the equality-based manifest embodiment. In new claims 19 and 35, each selected camera cut retains the same ordered transition from the first source camera to the second source camera at a mate camera-switch timing different from the reference camera-switch timing, with temporally corresponding frames from different source cameras during the interval between those timings. The searched delivered manifest file is one of the recipient-associated timing-choice manifest files generated, delivered and recorded by the claimed process. For each selected camera cut, the recipient-associated combination and the detected combination represented by the reconstructed manifest file include the same reference or mate camera-switch timing, with at least one mate camera-switch timing different from the corresponding reference timing. This complete chain from production to attribution is disclosed by original claim 1 read with page 12, line 16, through page 13, line 4, by pages 14-16, by original claims 11-15 and 17 and by Examples 1 and 3-5. The same ordered transition and the different-camera interval are concretely illustrated by Example 2.

As to the production side, the structured list of edit instructions with per-cut out-point and in-point time codes and source-camera identifiers is described at page 16, lines 18-28, and at page 18, lines 13-17. The direction-neutral mate-generation algorithm appears in Example 1 at pages 31-32. The metacode alternates between adding and subtracting time, assigns a variation of 1 or -1 and applies the variation to the camera-cut timestamp. Example 2 at pages 33-35, with Tables 1 and 2, provides the concrete delayed species by extending Cut 2, delaying the start of Cut 3 by ten frames and restoring synchronization from Cut 4 onward, thereby supporting the later-timing and subsequent-resynchronization fallbacks of new claims 20 and 36. The complete direction-neutral paired-entry and intervening different-camera relationship of claims 19 and 35 is supported by the direction-neutral algorithmic disclosure and original claim 1 read with the concrete relationship illustrated in Example 2. Variation at a plurality of selected camera cuts and generation of additional mates and manifest files are described at page 16, lines 1-17.

As to distribution and the record of associations, the chunk segmentation, content delivery network distribution with adaptive streaming, manifest tailoring to devices or network conditions, progressive mixing of reference and mate chunks and progressive assignment of individualized manifest files recited together in new claim 26 are described at page 8, lines 9-21, at page 14, lines 10-21, at page 18, line 23, through page 19, line 12, at page 29, lines 19-27, and at page 40, lines 5-7. Method 200 describes segmenting step 240 enabling adaptive streaming across varying network conditions and device capabilities and generation step 250 producing manifest files tailored to end user devices and network conditions. Original claims 11 and 12 recite the adaptive-delivery and progressive-mixing relationships in claim form. The ledger identifying which end user received which manifest file appears at page 8, lines 15-21, and at page 14, lines 17-21. See also original claims 13 and 14.

As to the single-recipient detection side, including monitor-side method claim 30, the operations of receiving the suspected unauthorized distribution, applying a scene-change detection algorithm to identify time codes of camera cuts, building one or more reconstructed manifest files from the identified time codes and searching the ledger to identify the associated recipient are described at page 8, lines 15-21, at page 14, line 22, through page 15, and in method form at page 30, lines 1-10, describing the ledger recording step 270, the monitoring step 280 detecting scene changes and devising time codes and the searching step 290. Original claims 15, 16 and 17 recite the same operations in claim form. Claim 30 identifies the recipient associated with the delivered manifest file representing the same cut-by-cut timing-choice combination as the reconstructed manifest file, including at least one detected different mate timing, and does not require identity of complete manifest syntax or metadata.

Independent claim 33 and dependent claim 34 address multi-contributor attribution without requiring the composite detected sequence to match the complete timing-choice sequence of one delivered manifest file. The colluding-redistribution disclosure at pages 23-26 states that portions from different copies remain traceable to their original distributions and that the ledger and the recorded chunk combinations identify the accounts involved. The probabilistic and Tardos disclosures at pages 24-26 describe identifying users who contributed to a composite pirate copy. The segmented-Tardos disclosure at pages 25-26 applies fingerprints to content segments to localize pirated segments and colluders. Read with the recipient-associated manifest combinations of pages 14-19, with the camera-cut monitoring and searching operations and with Example 2, these passages connect the collusion processing to sequences of chunk selections representing reference or mate timings of the same ordered transitions, to the intervening interval of temporally corresponding different-camera frames and to the determination, from each detected time code, whether the corresponding cut represents the reference or the mate timing. Generic Tardos processing does not by itself supply that timing-choice structure.

As to the remaining claims, the perceptual hash and fuzzy and sliding-window comparisons of new claims 24, 25, 37 and 38 are described at page 30, lines 21-28, in Example 5 at pages 40-42 and in original claims 2 and 3. The Edit Decision List implementation of the structured list, new claim 21, is described at page 16, lines 18-25. Capture of live video from diverse viewpoints and recording of director-commanded real-time camera selections, new claim 22, is recited in original claim 16. The computing environment recited in the system preamble is described at page 18, lines 1-3, and in original claim 8.

New claim 20 is supported by Example 2 at pages 33-35, which illustrates a mate transition later than the corresponding reference transition and restoration of synchronization at a subsequent camera cut. New claim 23 is supported by page 16, lines 1-8, describing further camera cuts spawning additional mates and manifest files, read together with Example 2 at pages 33-35, which provides the concrete single-cut variation at the Cut 2 and Cut 3 boundary. New claim 27 is supported by page 15, lines 12-16, which states expressly that the reference and mate manifest files point to sets of chunks of equal duration, together with Example 3 at pages 36-38, which shows the respective manifest files selecting different chunks at corresponding positions. New claim 28 is supported by the corresponding reference and mate chunk positions of Example 3 read together with the source camera sequence and the different reference and mate switch timings of Example 2. New claims 29 and 31 are supported by the delivered-manifest and reconstructed-manifest relationship at pages 14-15, by the specific chunks referenced at corresponding manifest positions in Example 3 at pages 36-38 and by the disclosure at page 19, lines 20-28, that a unique combination of chunks can be matched to a specific manifest file distributed to a particular spectator or group. New claim 32 is supported by the delivered-manifest and mate-chunk disclosure at pages 14-15 and in Example 3, read with the retained ordered transition at different timings and the intervening different-camera frames of Example 2. New claim 34 is supported by the segmented-Tardos disclosure at pages 25-26, read with the passages cited for claim 33. New claim 36 is supported by Example 2 at pages 33-35 in method-compatible production operations. For claims 23, 28, 29, 31, 32, 33 and 34, the recited relationships are shown by the cited passages read together rather than by a single passage.